

Project Design Phase
Problem – Solution Fit Template

Date	15 February 2025
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greengospace)	
Focus on J&P, tap into BE, understand RC	3. TRIGGERS What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure → confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	
Identify strong TR & EM				

Filled Canvas — Credit Card Approval Prediction

1. CUSTOMER SEGMENT(S) <i>Who is your customer?</i> - Retail & digital banks and fintech card issuers - Credit risk / underwriting teams within those institutions - Loan officers and credit analysts who review applications - Compliance & fair-lending officers who audit decisions	2. JOBS-TO-BE-DONE / PROBLEMS <i>Which jobs or problems do you address?</i> - Decide quickly whether to approve or decline a credit card application - Minimize default and credit-loss exposure from bad approvals - Reduce manual review time and inconsistent human judgment - Stay compliant with fair-lending regulations while scaling volume	3. TRIGGERS <i>What triggers customers to act?</i> - A new application arrives and needs a same-day decision - Rising default rates push leadership to demand better risk models - Regulators or auditors flag inconsistent manual decisions - Competitors launch instant-approval card products
6. CUSTOMER CONSTRAINTS <i>What limits their choice of solutions?</i> - Must comply with fair-lending / anti-discrimination law (e.g. ECOA) - Need explainable, auditable decisions, not a pure black box - Limited budget and IT capacity to integrate new scoring systems - Data privacy rules restrict which applicant data can be used	9. PROBLEM ROOT CAUSE <i>What is the real reason this problem exists?</i> - Traditional bureau scorecards are static and slow to reflect new risk patterns - Manual underwriting rules don't scale and vary reviewer to reviewer - Result: too many risky approvals (losses) or too many good customers declined (lost revenue)	5. AVAILABLE SOLUTIONS <i>What have they tried? Pros & cons?</i> - FICO / bureau credit scores — reliable but generic, not tailored to issuer's portfolio - Manual underwriting rules — explainable but slow and inconsistent - Legacy logistic-regression scorecards — auditable but low accuracy - Third-party approval APIs — fast but costly and low transparency
7. BEHAVIOUR <i>What do customers do to get the job done?</i> - Pull the applicant's credit bureau report and internal transaction history - Apply rule-based checks (income, existing debt, utilization) before escalation - Escalate borderline applications to a senior underwriter for manual review	10. YOUR SOLUTION <i>Current or proposed solution</i> - An ML model that scores each application on approval / default likelihood using applicant and bureau data - Automates clear-cut approvals and declines instantly, routes only borderline cases to human review - Includes an explainability layer so	4. EMOTIONS: BEFORE / AFTER <i>How do they feel before and after?</i> - Before: anxious about hidden risk, overwhelmed by backlog, unsure decisions are consistent - After: confident in a defensible, data-driven decision; in control of portfolio risk and turnaround

	decisions can be audited and defended to regulators	
8.1. CHANNELS — ONLINE <i>Online actions taken</i> • Internal loan/credit processing system used to submit and track applications • Bureau data APIs (Experian, Equifax, TransUnion) queried for credit history • Internal risk dashboards where analysts review model scores	8.2. CHANNELS — OFFLINE <i>Offline actions taken</i> • In-branch application intake and document verification • Phone follow-ups with applicants for missing information • Underwriting committee meetings for high-value or disputed cases	

Purpose / Vision: Use predictive modeling to make credit card approval decisions faster, fairer, and more accurate — reducing losses while approving more creditworthy customers.

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>